

Program : Diploma in Civil Engineering and Planning	
Course Code: 5391	Course Title: Construction Planning and Management
Semester: 5	Credits: No Credit
Course Category: Common Course	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To know the principle involved in construction management
- To understand the contract management and associated labour laws.
- To prepare and understand the principle involved in site layout.
- To know the procedure for scheduling of various activities in construction project.
- To familiarize with labour laws, procedure for arbitration, settlements.
- To know different safety measures in construction projects.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Stages of Construction		Building construction and construction materials	3

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	Identify principles involved in construction management and explain features of construction organization	15	Understanding
CO2	Apply scheduling techniques in project planning	16	Applying
CO3	Understand contracting procedure and familiarize various equipment involved in construction.	14	Understanding

CO4	Understand the basics of Construction Resource Management and Safety Management	13	Understanding
	Series Test	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1						3	
CO2						3	
CO3						3	
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify principles involved in construction management and explain features of construction organization		
M1.01	Identify the functions and importance of management	5	Remembering
M1.02	Understand the concept of construction management	4	Understanding
M1.03	Explain the basic principles and forms of construction organization	3	Understanding
M1.04	Explain the preparation of site layout	3	Understanding

Contents:

Definition and objective of management, F.W. Taylor's and Henry Fayol's Principles of Management, Functions of Manager, functions of management, management of project coordination-project management team, stages of project, phases of project management work breakdown structure.

Construction team, construction agencies, stages of construction, Details to be collected during planning and preparation of detailed project report, resources of construction, detailed functions of construction management, persons required on site.

Management organization, principles of organization, forms of organization-project organization-types, methods of execution of work

Job-layout, requirement of site layout, factors affecting job layout, Preparation of site layout

CO2	Apply scheduling techniques in project planning		
M2.01	Use bar charts to show the activities in a construction work	3	Applying
M2.02	Identify the terms related with network for the given construction project	4	Understanding
M2.03	Schedule network for the given construction project	5	Applying
M2.04	Describe material management.	4	Understanding
	Series Test - I	1	

Contents:

Identifying broad activities in construction work & allotting time period to it, Methods of Scheduling, Development of bar charts, Merits & limitations of bar chart.

Elements of Network: Event, activity, dummy activities, Precautions in drawing Network, Numbering the events.

CPM networks, activity time estimate, start and finish time of activity, project duration. Floats: Types of Floats-Free, independent and total floats, critical activities and critical path. Outline PERT networks

Purpose of crashing a network, Normal Time and Cost, Crash Time and Cost, Cost slope, Optimization of cost and duration.

Introduction to BIM- Management Software

Material Management- Ordering cost, inventory carrying cost, Economic Order Quantity

Store management, various records related to store management, inventory control by ABC, VED technique, Introduction to material procurement through portals (e.g., www.inampro.nic.in)

CO3	Understand contracting procedure and familiarize various equipment involved in construction.		
M3.01	Describe contracting procedure	4	Understanding
M3.02	Explain the objectives and importance of quality control	3	Remembering
M3.03	Explain uses and working of different equipment used in construction projects	4	Understanding
M3.04	Explain how to choose appropriate equipment used in construction projects	3	Understanding
Contents: Types of Construction contracts. Contract documents, specifications, general and special conditions. Contract Management, procedures involved in arbitration and settlement (Introduction only) Quality control: – Need and objectives of quality control – Role of specification, supervision, check lists, inspection and sampling & testing of materials in quality control. Identify different modern equipment used in construction projects. -bull dozer, angle dozer, excavator, power shovel, forklift, tower crane, back hoe, back hoe loader, gantry crane, slip form paver, grader, TBM etc. State the specific use of various construction equipment. Introduction to Equipment Economics: Owning and Operating Costs, uses of Machines Vs Manual power, forecasting equipment requirement, Factors for selection of equipment.			
CO4	Understand the basics of Construction Resource Management and Safety Management		
M4.01	Understand the necessity of Construction Resource Management	4	Remembering
M4.02	Explain the causes of accidents and safety practices at construction site	4	Understanding
M4.03	Explain precautions to be taken in handling hazardous material and describe role of supervisor/ Engineer in ensuring safety	2	Understanding
M4.04	Describe Labour Laws and Acts	3	Understanding
	Series Test - II	1	
Contents: Resource Management: Definition – Need for resource management – Optimum utilization of resources- finance, materials, machinery, human resources - Resource planning – Resource levelling and its objectives Construction safety: Basic principles of safety - Major causes of accidents at project site - Effects of accidents - Safety practices at construction site - Excavation, Working at Heights, Fire, Underwater, Marshlands etc.-			

Precautions in handling hazardous materials - Occupational hazard - Role of supervisor/ Engineer in ensuring safety at construction site -
Labour Laws and Acts pertaining to Civil construction activities (Introduction)

Text/Reference:

T/R	Book Title/Author
T1	Sharma S C and Deodhar S V, Construction Engineering and Management, Khanna Book Publishing, New Delhi
R1	Gahlot,P.S. and Dhir, B.M Construction planning and managemen New Age International (P) Ltd. Publishers, New Delhi.
R2	Shrivastava, U.K., Construction planning and management, Galgotia Publication Pvt Ltd. New Delhi
R3	Mantri, S., The A to Z of Practical Building Construction and its Management, Satya Prakashan New Delhi
R4	Khanna, O.P., Industrial Engineering and management, Dhanpat Rai New Delhi
R5	Punmia, B.C. and Khandelwal, K.K., Project Planning and Controlling with PERT and CPM,
R6	Sengupta, B., Guha H., Construction Management and Planning, Tata-McGrawHill
R7	Harpal, Singh, Construction Management and accounts, Mc-Graw Hill.
R8	Sharma, S.C., Industrial Engineering and Management, Khanna Publications, New Delhi
R9	B C Punmia, “ Project Planning and control with PERT and CPM”, Laxmi Publications

Online Resources:

Sl.No	Website Link
1	www.inampro.nic.in
2	https://www.mindtools.com/pages/article/critical-path-analysis.htm
3	https://etenders.kerala.gov.in/nicgep/app
4	https://theconstructor.org/construction/heavy-construction-equipment-types/26305/
5	https://ehs.princeton.edu/workplace-construction/construction-safety